Task 1 – ML Linear Regression

California Housing Price Prediction

Artificial Intelligence & Machine Learning

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This report summarizes the implementation and evaluation of a Linear Regression model using the California Housing dataset.

1. Introduction

The main objective of this project was to understand the complete machine learning workflow using a simple regression problem. A Linear Regression model was trained on the California Housing dataset to predict median house prices using multiple housing-related features.

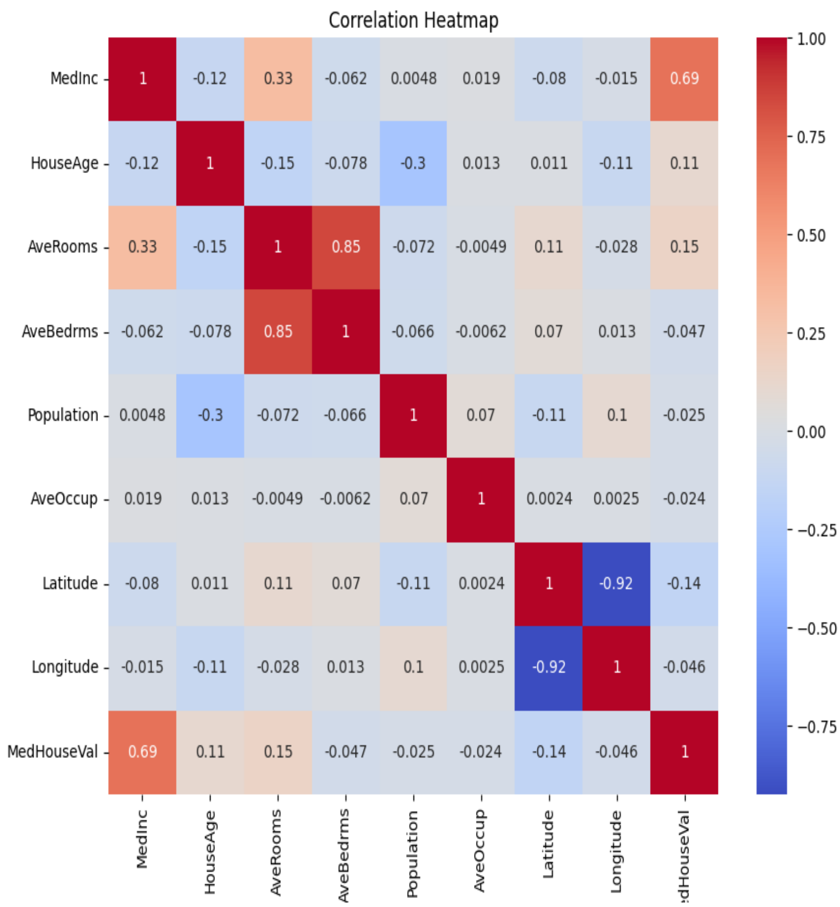
The project included data exploration, visualization, model training, evaluation, and saving the trained model for future use.

2. Dataset Overview

The California Housing dataset contains information about housing districts in California. Important features used in this project include Median Income, House Age, Average Rooms, Population, Latitude, Longitude, and Median House Value.

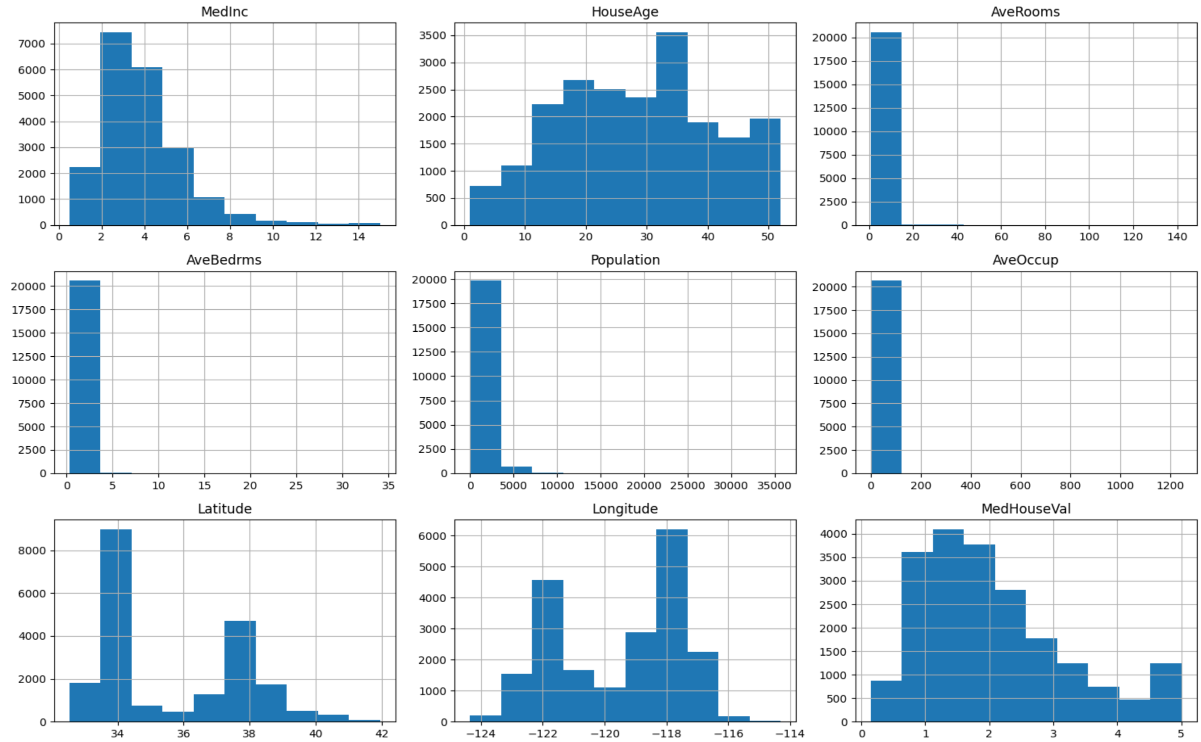
3. Correlation Analysis

A correlation heatmap was used to understand relationships between different variables. Median Income showed the strongest positive correlation with house prices, while Latitude and Longitude showed a strong negative relationship.



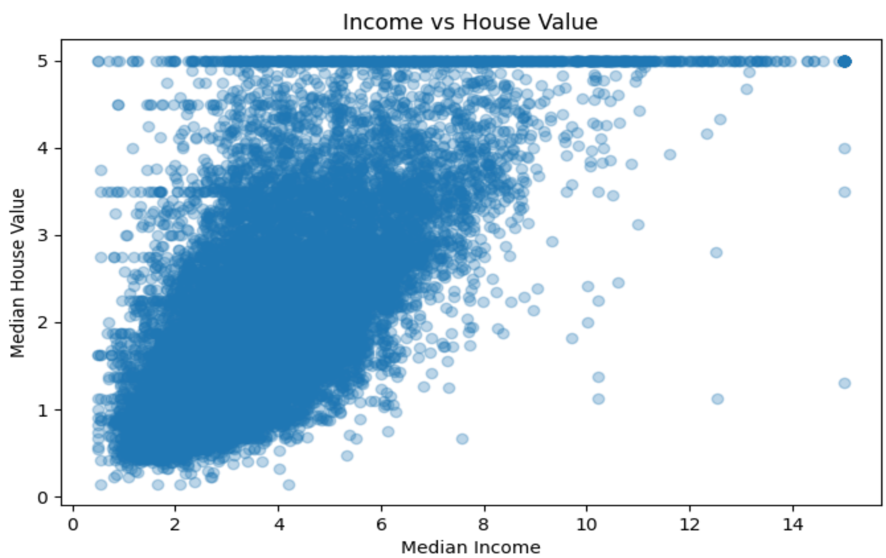
4. Feature Distribution

Histograms were generated to observe the distribution of all important features. Some variables such as Population and Average Occupancy were highly skewed, while Median Income and House Age had comparatively smoother distributions.



5. Income vs House Value

The scatter plot below shows a positive relationship between median income and median house value. As the income level increases, house prices also tend to increase.



6. Model Training & Evaluation

The dataset was divided into training and testing sets using an 80-20 split. A Linear Regression model from Scikit-learn was trained using the training dataset. After training, predictions were generated on the test data and evaluated using different regression metrics.

Metric	Result
MAE	0.5332
RMSE	0.7456
R^2 Score	0.5758

7. Possible Improvements

The model performance can be improved further by using feature engineering, data scaling, outlier handling, or advanced regression algorithms such as Random Forest Regressor or XGBoost.

8. Conclusion

This project helped in understanding the complete workflow of a machine learning regression model, starting from data exploration to final evaluation. The Linear Regression model was able to capture the relationship between housing features and house prices with reasonable accuracy.